

# AI Personal Assistant Complete User Guide

Visual Feature Walkthrough with Real Screen Previews, Multi-Provider API Setup Manual (OpenRouter, Gemini, Groq, DeepSeek), Model Hub Navigation, and Autonomous Scheduling.

✦ CORE SYSTEM CAPABILITIES

- Hermes Multi-Provider API Architecture
- 450+ Cloud Models Directory & Instant Search
- Free Tier LLM Hub (Nemotron, Ling, Qwen)
- On-Device Deterministic NLP Action Parser
- Autonomous AlarmManager Exact Reminders
- Local Room Database (Tasks, Notes, Chats)

PACKAGE / APPLICATION ID

com.mitaoe.shridhar202401040197.aiassistant

APPLICATION VERSION

1.0 (Release / Debug)

TARGET OS COMPATIBILITY

Android 8.0 to Android 14+

UI DESIGN LANGUAGE

Obsidian Cybernetic (Dark Neon)

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System Architecture: The Dual-Engine Advantage

The AI Personal Assistant is uniquely engineered to provide uninterrupted utility regardless of your network connectivity or API quota status:

1. Local Deterministic Engine (Offline)

Powered by `NaturalLanguageActionParser.java`. Runs 100% locally on your phone with **zero API key and zero internet required**. It automatically intercepts commands like *"Remind me to call professor tomorrow at 3pm"*, saves them to Room SQLite, and sets exact hardware alarms.

2. Cloud Generative LLM Engine (Online)

Powered by `AIService.java` with Hermes-style multi-provider support. Allows you to plug in API keys from **OpenRouter, Google Gemini, Groq, DeepSeek, or OpenAI** to gain access to 450+ conversational reasoning models with live token streaming.

 **Privacy by Design:** The application stores all notes, tasks, alarms, and chat histories inside an encrypted local Room SQLite database on your device. Your personal data is never sent to third-party tracking or analytics servers.

# 1. How to Add API Keys & Where to Find Them

To unlock online cloud conversational intelligence, you can connect one or more API providers. The app supports 9 providers with zero setup lock-in.

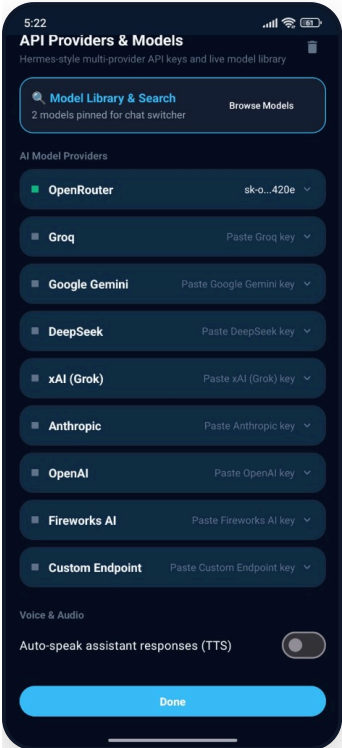


Fig 1.1: API Providers & Models Dialog

## Where to Find Free & Supported APIs:

| Provider                 | Website / Portal                          | Cost & Recommended Models                                                                       |
|--------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------|
| OpenRouter (Recommended) | <code>openrouter.ai/keys</code>           | <b>100% FREE TIER</b><br>Access to 20+ free models (Nemotron 3, Ling 3.0, Llama 3.3, Qwen 27B). |
| Google Gemini            | <code>aistudio.google.com</code>          | <b>FREE GENERATIVE QUOTA</b><br>Gemini 1.5 Flash (1M token context, high speed).                |
| Groq                     | <code>console.groq.com/keys</code>        | <b>FREE TIER</b><br>Llama 3.3 70B with ultra-fast 500+ tokens/sec inference.                    |
| DeepSeek                 | <code>platform.deepseek.com</code>        | <b>LOW COST PAY-AS-GO</b><br>DeepSeek-V3 and DeepSeek-R1 reasoning.                             |
| OpenAI                   | <code>platform.openai.com/api-keys</code> | <b>COMMERCIAL API</b><br>GPT-4o and GPT-4o-mini.                                                |
| Local / Custom           | <code>http://localhost:11434/v1</code>    | <b>SELF-HOSTED</b><br>Local Ollama, vLLM, or LM Studio running on local Wi-Fi.                  |

## Step-by-Step Guide: How to Add Your API Key in the App

- 1

**Open Settings:** Tap the **Gear** (⚙️) icon on the Dashboard or in the Chat top bar.
- 2

**Select Your Provider:** In the *AI Model Providers* list, tap on your provider (e.g., **OpenRouter** or **Google Gemini**).
- 3

**Paste Your Key:** Enter or paste your key into the text field (e.g. `sk-or-v1- ...`).
- 4

**Verify Green Indicator:** A solid **green badge** ■ appears next to the provider name confirming your key is saved.
- 5

**(Optional) Enable Voice TTS:** Toggle "*Auto-speak assistant responses (TTS)*" to hear responses read aloud.
- 6

**Save:** Tap **Done** at the bottom. Your AI Engine is immediately **Online & Ready!**

## 2. Model Library & Search Hub

The app includes an integrated Model Discovery Catalog allowing you to explore over **450+ live models** or quickly pick from **20+ completely free** models.

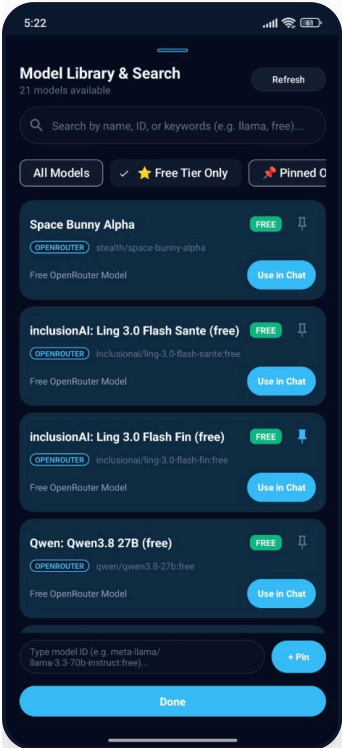


Fig 2.1: Free Tier Only Filter

### Navigating the Model Library

To open the Model Hub, tap **Model Library & Search** from the Settings screen or select **Model Library & Hub** from the left navigation drawer.

#### ★ Filter: Free Tier Only

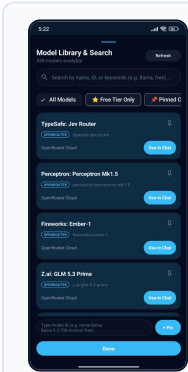
Tap the ★ **Free Tier Only** chip at the top to filter models that require **\$0.00 credits** on OpenRouter:

- **NVIDIA: Nemotron 3 Ultra (free)**: Ultra-capable general reasoning model.
- **inclusionAI: Ling 3.0 Flash Fin / Sante (free)**: Lightweight, lightning-fast response times.
- **Qwen: Qwen3.8 27B (free)**: High-parameter multilingual and mathematical model.
- **Space Bunny Alpha**: Creative synthesis and rapid summarization.

#### 📌 Pinning Models & One-Tap Activation

- **Use in Chat**: Instantly makes the selected model your primary conversational engine and switches back to the chat screen.
- **Pin Icon** (📌): Pins the model to your *Quick Switcher* bottom sheet so you can toggle between it and other models without re-opening the library.
- **Manual Model Pin**: At the bottom, you can paste any custom model identifier (e.g. `meta-llama/llama-3.3-70b-instruct:free`) and tap **+ Pin**.

🔍 **Live Search**: Type any keyword in the search bar (e.g., *"llama"*, *"code"*, *"qwen"*) to filter the catalog in real-time with sub-millisecond responsiveness.



### Full Cloud Directory (458+ Models Available)

Switching to the **All Models** tab provides live access to cutting-edge models from TypeSafe, Fireworks AI, Z.ai GLM, Perceptron, Mistral, and Anthropic. Pinned items are saved in Room DB across sessions.

### 3. AI Chat Command Center & Quick Model Switcher

The Chat screen combines intelligent generative conversation with seamless multi-model switching, speech-to-text dictation, and speech synthesis.

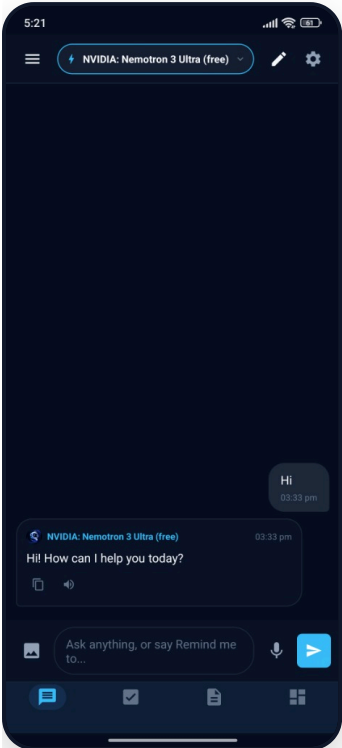
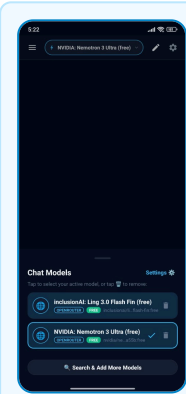


Fig 3.1: Active Chat Screen

#### Core Chat Interface Elements

- **Active Model Header Chip:** The blue pill at the top displays the active model (e.g., **NVIDIA: Nemotron 3 Ultra (free)**). Tap it to open the quick switcher!
- **Pencil & Gear Icons:** Tap the **Pencil** (✎) to start a clean new chat session; tap the **Gear** (⚙️) to jump directly to API configuration.
- **Message Action Tools:** Every AI response includes two quick action buttons:
  - **Copy Icon** (📄): Copies the complete formatted markdown response to your system clipboard.
  - **Speaker Icon** (🔊): Triggers text-to-speech to read the assistant's answer out loud.
- **Multi-Modal Attachments:** Tap the **Gallery Icon** (🖼️) to attach images for visual analysis.
- **Voice Microphone** (🎤): Tap to dictate your questions or commands hands-free via Android speech recognition.

#### In-Chat Quick Model Switcher



##### Frictionless Switching (Fig 3.2)

- Tapping the top header chip opens the *Chat Models* bottom sheet:
- **Tap to Select:** Tap any pinned model to switch immediately. A checkmark (✓) indicates the active model.
  - **Trash Can** (🗑️): Remove a model from your pinned list.
  - **Search & Add:** Opens the full 450+ model catalog to pin new ones.

## 4. Cyber Dashboard & Morning Audio Briefing

The Dashboard is the dynamic headquarters of your day, providing real-time metric aggregates, audio executive summaries, and schedule tracking.

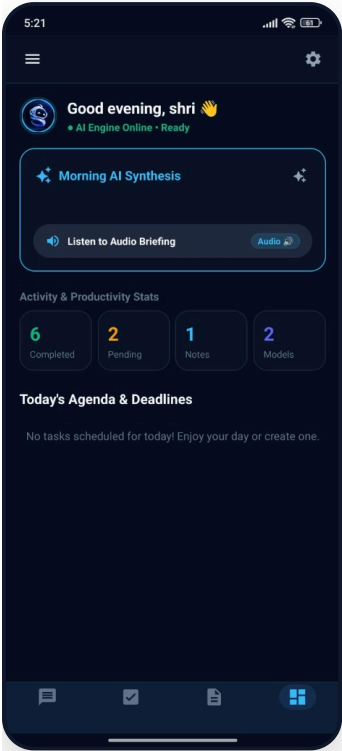


Fig 4.1: Cyber Dashboard Overview

### Dashboard Components Breakdown

#### 1. Personalized Greeting & Engine Status

- Displays "Good evening, shri 🖐️", dynamically updating based on system time and authenticated user profile.
- **AI Engine Online • Ready**: Real-time heartbeat checking whether active provider credentials and local databases are operational.

#### 2. Morning AI Synthesis & Audio Briefing

- The **Morning AI Synthesis** card uses LLM reasoning to summarize pending tasks, notes, and calendar deadlines into a 30-second daily briefing.
- Tap "**Listen to Audio Briefing**" 🔊 to listen to your morning schedule read aloud via Android Text-To-Speech while getting ready.

#### 3. Activity & Productivity Stats

Live SQLite counters tracking your organization progress:

|           |         |       |        |
|-----------|---------|-------|--------|
| 6         | 2       | 1     | 2      |
| Completed | Pending | Notes | Models |

#### 4. Today's Agenda & Deadlines

Automatically surfaces tasks due before midnight today. If no tasks remain, displays "No tasks scheduled for today! Enjoy your day or create one."

## 5. AI Workspace & Navigation Drawer

Tapping the top-left hamburger menu (☰) slides open the **AI Workspace Drawer**, serving as your personal account and session manager.

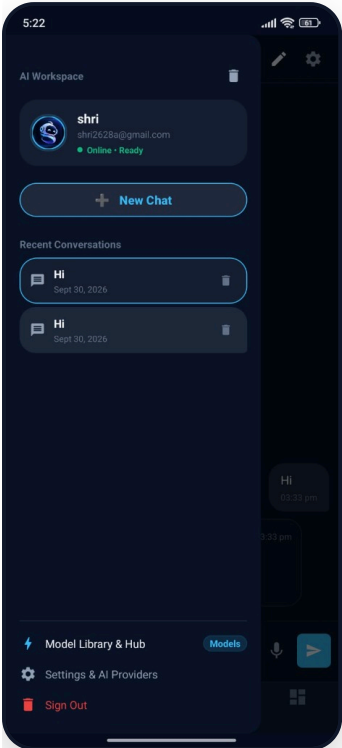


Fig 5.1: AI Workspace Drawer

### Drawer Features & Controls

#### User Profile Card

- Displays custom avatar with neon obsidian cyber ring.
- Shows registered user name ( `shri` ) and email ( `shri2628a@gmail.com` ).
- **Online • Ready**: Confirms local database and cloud API connectivity.

#### + New Chat Button

Tapping the large pill button initiates an immediate fresh conversation thread, saving the current conversation to history.

#### Recent Conversations History

- Shows chronological list of past chat sessions (e.g. *"Hi - Sept 30, 2026"*).
- Tap any conversation card to reopen and restore full message history.
- Tap the **Trash Can** (🗑️) icon on any session to delete that specific transcript from Room SQLite.

#### Shortcuts & Account Management

- ⚡ **Model Library & Hub**: Direct jump to the 450+ model catalog.
- ⚙️ **Settings & AI Providers**: Fast access to API key management.
- 🚪 **Sign Out (Red)**: Securely ends the current session and returns to the authentication screen.

## 6. Task Matrix & Autonomous Alarms

The **Tasks Matrix** (`TasksFragment.java`) delivers robust schedule planning with priority classification, checkbox states, and system-level alarms.

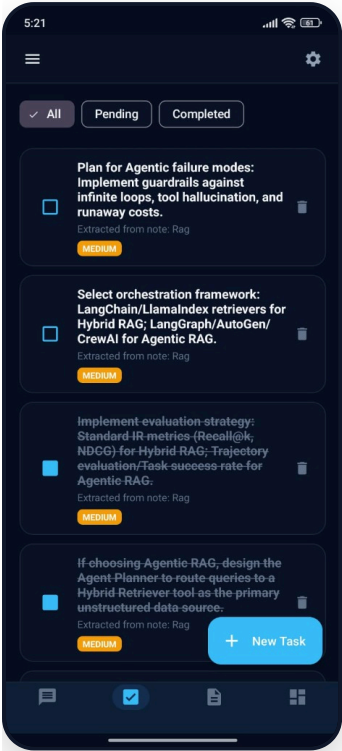


Fig 6.1: Task Matrix with Extracted Tasks

### Managing Tasks in the Matrix

#### 1. Filter Tabs: All / Pending / Completed

Quickly toggle between ✓ **All** (shows full list), **Pending** (focuses on unfinished work), and **Completed** (shows strikethrough accomplished items).

#### 2. Priority Badges & Completion Checkboxes

- **Interactive Checkboxes:** Tapping the square checkbox immediately toggles completion status, striking through text and updating dashboard stats.
- **Priority Tags:** Tasks are categorized with color-coded tags: **HIGH** (red), **MEDIUM** (amber), **LOW** (green).

#### 3. AI Smart Task Extraction from Notes!

Notice the subtitle on the tasks in Fig 6.1:  
`Extracted from note: Rag`  
The assistant allows you to extract actionable todo items from any research note or lecture with a single tap, automatically populating the task matrix!

#### 4. + New Task Floating Action Button

Tap the bright cyan **+ New Task** button at bottom-right to open the creation dialog, set deadlines, and configure exact notification alarms.

## 7. Knowledge Vault (Notes) & AI Extraction Tools

The **Notes** screen acts as your intellectual repository, enhanced with one-touch AI processing tools for research synthesis.

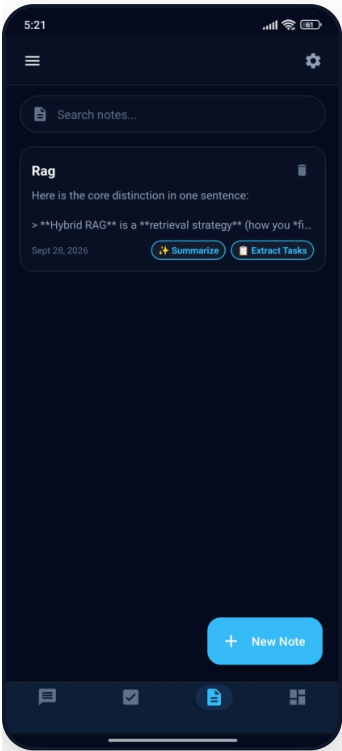


Fig 7.1: Knowledge Vault with AI Tools

### Supercharged AI Note Operations

#### 1. One-Tap AI Summarize

Tapping the ✨ **Summarize** chip on any note feeds its content into the active LLM engine to extract key executive takeaways in 3 concise bullet points.

#### 2. One-Tap Extract Tasks

Tapping 📄 **Extract Tasks** automatically parses the note's action items, architectural plans, and deliverables, and converts them directly into records in your **Task Matrix**!

#### 3. Live Full-Text Search

Use the top **Search notes...** bar to instantly filter across note titles, body text, and creation dates.

#### + 4. Rich Note Creation (+ New Note)

Tap the floating cyan **+ New Note** button to compose research documentation, lecture notes, or code snippets with markdown preview.

#### 5. Clean Management & Deletion

Tap the trash icon on the top-right of any note card to remove it permanently from local Room database storage.

## 8. Natural Language Voice Commands Cheat Sheet

Tap the **Microphone** (🎤) on any screen and speak these natural phrases to execute actions instantly:

| Target Action      | Voice / Natural Language Command                   | Resulting App Behavior                                            |
|--------------------|----------------------------------------------------|-------------------------------------------------------------------|
| Create Urgent Task | "Add task submit lab journal tomorrow 10am urgent" | Creates HIGH priority task, sets due date, registers exact alarm. |
| Set Reminder       | "Remind me to attend seminar tonight at 8 PM"      | Parses "tonight 8 PM", sets exact alarm, confirms in chat.        |
| General Task       | "New task buy circuit components for IoT project"  | Creates standard task with MEDIUM priority in Task Matrix.        |
| Quick Note         | "Take a note: Hybrid RAG is a retrieval strategy"  | Saves note directly to Knowledge Vault with timestamp.            |
| Complex Query      | "Compare Agentic RAG vs Standard RAG architecture" | Routes query to active cloud LLM for full technical synthesis.    |

## Troubleshooting & Permissions Guide

### 🔔 Alarm / Notification Not Ringing?

- Open Android Settings → Apps → **AI Personal Assistant**.
- Set Battery Usage to **"Unrestricted"** so Android Doze Mode doesn't pause alarms.
- Ensure **"Allow notifications"** is toggled ON.

### 🎤 Speech Recognition Error?

- Go to App Permissions → **Microphone** → Select *"Allow only while using the app"*.
- Verify Google Speech Services / Voice typing is enabled in Android Keyboard settings.

⚙️ **Switching Back to Offline Mode:** If you run out of internet or API credits, the local deterministic engine continues to parse all tasks, notes, and reminders automatically without showing errors.